

Kafka 集群安装部署

简介

Kafka 是一款分布式的、去中心化的、高吞吐低延迟、订阅模式的消息队列系统，多用于大数据体系。

安装

注意：安装 Kafka 之前必须安装并部署好了 JDK 和 Zookeeper 环境

1. 【在 node1 操作】：下载并上传 Kafk 安装包

因为在服务器使用 wget 下载速度较慢，我直接在官网下载合适的安装包，上传到服务器上，后续都会放到实验仓库中



开始解压，解压完成之后，查看文件夹有没有该文件

```
[root@node1 server]# ll
总用量 8
drwxr-xr-x. 9 tomcat tomcat 4096 4月 22 21:06 apache-tomcat-8.5.100
drwxr-xr-x. 8 root root 157 4月 25 18:38 apache-zookeeper-3.8.4-bin
lrwxrwxrwx. 1 root root 28 4月 22 20:25 jdk -> /export/server/jdk1.8.0_491/
drwxr-xr-x. 8 root root 4096 4月 22 20:24 jdk1.8.0_491
drwxr-xr-x. 7 root root 105 3月 28 2024 kafka_2.13-3.6.2
lrwxrwxrwx. 1 tomcat tomcat 36 4月 22 21:09 tomcat8 -> /export/server/apache-tomcat-8.5.100
lrwxrwxrwx. 1 root root 41 4月 25 18:00 zookeeper -> /export/server/apache-zookeeper-3.8.4-bin
```

创建一个软链接

```
[root@node1 server]# ln -s kafka_2.13-3.6.2 /export/server/kafka
[root@node1 server]# ll
总用量 8
drwxr-xr-x. 9 tomcat tomcat 4096 4月 22 21:06 apache-tomcat-8.5.100
drwxr-xr-x. 8 root root 157 4月 25 18:38 apache-zookeeper-3.8.4-bin
lrwxrwxrwx. 1 root root 28 4月 22 20:25 jdk -> /export/server/jdk1.8.0_491/
drwxr-xr-x. 8 root root 4096 4月 22 20:24 jdk1.8.0_491
lrwxrwxrwx. 1 root root 16 4月 25 20:21 kafka -> kafka_2.13-3.6.2
drwxr-xr-x. 7 root root 105 3月 28 2024 kafka_2.13-3.6.2
lrwxrwxrwx. 1 tomcat tomcat 36 4月 22 21:09 tomcat8 -> /export/server/apache-tomcat-8.5.100
lrwxrwxrwx. 1 root root 41 4月 25 18:00 zookeeper -> /export/server/apache-zookeeper-3.8.4-bin
```

2. 【在 node1 操作】：进入 kafka 的 config 目录，进入 server.properties 进行参数修改

```
[root@node1 config]# ll
总用量 72
-rw-r--r-- 1 root root 906 3月 28 2024 connect-console-sink.properties
-rw-r--r-- 1 root root 909 3月 28 2024 connect-console-source.properties
-rw-r--r-- 1 root root 5475 3月 28 2024 connect-distributed.properties
-rw-r--r-- 1 root root 883 3月 28 2024 connect-file-sink.properties
-rw-r--r-- 1 root root 881 3月 28 2024 connect-file-source.properties
-rw-r--r-- 1 root root 2863 3月 28 2024 connect-log4j.properties
-rw-r--r-- 1 root root 2540 3月 28 2024 connect-mirror-maker.properties
-rw-r--r-- 1 root root 2262 3月 28 2024 connect-standalone.properties
-rw-r--r-- 1 root root 1221 3月 28 2024 consumer.properties
drwxr-xr-x. 2 root root 85 3月 28 2024 kraft
-rw-r--r-- 1 root root 4917 3月 28 2024 log4j.properties
-rw-r--r-- 1 root root 2065 3月 28 2024 producer.properties
-rw-r--r-- 1 root root 6896 3月 28 2024 server.properties
-rw-r--r-- 1 root root 1094 3月 28 2024 tools-log4j.properties
-rw-r--r-- 1 root root 1169 3月 28 2024 trogdor.conf
-rw-r--r-- 1 root root 1205 3月 28 2024 zookeeper.properties
```

①将 `broker.id` 改为 1

```
22
23 # The id of the broker. This must be set to a unique integer for each broker.
24 broker.id=0
25
```

②在 `listeners` 后面添加 `node1`

```
32 # EXAMPLE:
33 # listeners = PLAINTEXT://your.host.name:9092
34 listeners=PLAINTEXT://:9092
```

```
# listeners = PLAINTEXT://your.host.name:9092
listeners=PLAINTEXT://node1:9092
```

③创建 kafka 存放数据的文件夹

```
60
61 # A comma separated list of directories under which to store log files
62 log.dirs=/export/server/kafka/data
63
```

④在 `zookeeper.connect` 后添加服务器节点

```
125 zookeeper.connect=node1:2181,node2:2181,node3:2181
126
```

3. 【在 node1 上操作】：将 kafka 的安装包上传到 node2 和 node3 的 `/export/server` 目录

```
[root@node2 server]# ll
总用量 8
drwxr-xr-x. 9 tomcat tomcat 4096 4月 22 21:06 apache-tomcat-8.5.100
drwxr-xr-x. 8 root root 157 4月 25 18:38 apache-zookeeper-3.8.4-bin
lrwxrwxrwx. 1 root root 28 4月 22 20:25 jdk -> /export/server/jdk1.8.0_491/
drwxr-xr-x. 8 root root 4096 4月 22 20:24 jdk1.8.0_491
drwxr-xr-x. 8 root root 117 4月 25 20:32 kafka 2.13-3.6.2 上传成功
lrwxrwxrwx. 1 tomcat tomcat 36 4月 22 21:09 tomcat8 -> /export/server/apache-tomcat-8.5.100
lrwxrwxrwx. 1 root root 26 4月 25 18:31 zookeeper -> apache-zookeeper-3.8.4-bin
[root@node2 server]# ln -s kafka_2.13-3.6.2 /export/server/kafka
[root@node2 server]# ll
总用量 8
drwxr-xr-x. 9 tomcat tomcat 4096 4月 22 21:06 apache-tomcat-8.5.100
drwxr-xr-x. 8 root root 157 4月 25 18:38 apache-zookeeper-3.8.4-bin
lrwxrwxrwx. 1 root root 28 4月 22 20:25 jdk -> /export/server/jdk1.8.0_491/
drwxr-xr-x. 8 root root 4096 4月 22 20:24 jdk1.8.0_491
lrwxrwxrwx. 1 root root 16 4月 25 20:32 kafka -> kafka_2.13-3.6.2 软链接也已经创建
drwxr-xr-x. 8 root root 117 4月 25 20:32 kafka_2.13-3.6.2
lrwxrwxrwx. 1 tomcat tomcat 36 4月 22 21:09 tomcat8 -> /export/server/apache-tomcat-8.5.100
lrwxrwxrwx. 1 root root 26 4月 25 18:31 zookeeper -> apache-zookeeper-3.8.4-bin
[root@node2 server]#
```

```
[root@node3 server]# ll
总用量 8
drwxr-xr-x. 9 tomcat tomcat 4096 4月 22 21:06 apache-tomcat-8.5.100
drwxr-xr-x. 8 root root 157 4月 25 18:38 apache-zookeeper-3.8.4-bin
lrwxrwxrwx. 1 root root 28 4月 22 20:25 jdk -> /export/server/jdk1.8.0_491/
drwxr-xr-x. 8 root root 4096 4月 22 20:24 jdk1.8.0_491
drwxr-xr-x. 8 root root 117 4月 25 20:32 kafka 2.13-3.6.2 上传成功
lrwxrwxrwx. 1 tomcat tomcat 36 4月 22 21:09 tomcat8 -> /export/server/apache-tomcat-8.5.100
lrwxrwxrwx. 1 root root 26 4月 25 18:31 zookeeper -> apache-zookeeper-3.8.4-bin
[root@node3 server]# ln -s kafka_2.13-3.6.2 /export/server/kafka
[root@node3 server]# ll
总用量 8
drwxr-xr-x. 9 tomcat tomcat 4096 4月 22 21:06 apache-tomcat-8.5.100
drwxr-xr-x. 8 root root 157 4月 25 18:38 apache-zookeeper-3.8.4-bin
lrwxrwxrwx. 1 root root 28 4月 22 20:25 jdk -> /export/server/jdk1.8.0_491/
drwxr-xr-x. 8 root root 4096 4月 22 20:24 jdk1.8.0_491
lrwxrwxrwx. 1 root root 16 4月 25 20:45 kafka -> kafka_2.13-3.6.2 软链接也已经创建
drwxr-xr-x. 8 root root 117 4月 25 20:32 kafka_2.13-3.6.2
lrwxrwxrwx. 1 tomcat tomcat 36 4月 22 21:09 tomcat8 -> /export/server/apache-tomcat-8.5.100
lrwxrwxrwx. 1 root root 26 4月 25 18:31 zookeeper -> apache-zookeeper-3.8.4-bin
[root@node3 server]#
```

4. 【在 node2 和 node3 上操作】：和操作 2 一样，进入 kafka 的 config 目录，

进入 server.properties 进行参数修改

将 broker.id 改为 2，listeners 添加 node2

```
# The id of the broker. This must be set to a unique integer for each broker.
broker.id=2

#
#   listeners = PLAINTEXT://your.host.name:9092
listeners=PLAINTEXT://node2:9092
```

将 broker.id 改为 3，listeners 添加 node3

```
# The id of the broker. This must be set to a unique integer for each broker.
broker.id=3

#
#   listeners = PLAINTEXT://your.host.name:9092
listeners=PLAINTEXT://node3:9092
```

5. 启动 Kafka

注意：启动 kafka 之前，一定要先启动 Zookeeper

方式 1——【前台启动】：在 node1、2、3 上分别执行 `/export/server/kafka/bin/kafka-server-start.sh /export/server/kafka/config/server.properties`

这是启动时要带的参数

在 /kafka/bin 目录下有一个 `kafka-server-start.sh` 文件，启动该文件就可以启动 Kafka

```
-rwxr-xr-x 1 root root 873 3月 28 2024 kafka-metadata-shell.sh
-rwxr-xr-x 1 root root 862 3月 28 2024 kafka-mirror-maker.sh
-rwxr-xr-x 1 root root 959 3月 28 2024 kafka-producer-perf-test.sh
-rwxr-xr-x 1 root root 874 3月 28 2024 kafka-reassign-partitions.sh
-rwxr-xr-x 1 root root 885 3月 28 2024 kafka-replica-verification.sh
-rwxr-xr-x 1 root root 10884 3月 28 2024 kafka-run-class.sh
-rwxr-xr-x 1 root root 1376 3月 28 2024 kafka-server-start.sh
-rwxr-xr-x 1 root root 1361 3月 28 2024 kafka-server-stop.sh
-rwxr-xr-x 1 root root 860 3月 28 2024 kafka-storage.sh
-rwxr-xr-x 1 root root 956 3月 28 2024 kafka-streams-application-reset.sh
-rwxr-xr-x 1 root root 863 3月 28 2024 kafka-topics.sh
```

执行命令（三台都要执行）

```
[root@node1 bin]# /export/server/kafka/bin/kafka-server-start.sh /export/server/kafka/config/server.properties
[2026-04-25 21:03:28,999] INFO Registered kafka:type=kafka.Log4jController MBean (kafka.utils.Log4jControllerRegistration$)
[2026-04-25 21:03:29,540] INFO Setting -D jdk.tls.rejectClientInitiatedRenegotiation=true to disable client-initiated TLS renegotiation (org.apache.zookeeper.common.X509Util)
[2026-04-25 21:03:29,699] INFO Registered signal handlers for TERM, INT, HUP (org.apache.kafka.common.utils.LoggingSignalHandler)
[2026-04-25 21:03:29,701] INFO starting kafka.server.KafkaServer
[2026-04-25 21:03:29,782] INFO Connecting to zookeeper on node1:2181,node2:2181,node3:2181 (kafka.server.KafkaServer)
[2026-04-25 21:03:29,718] INFO (ZooKeeperClient:Kafka server) Initializing a new session to node1:2181,node2:2181,node3:2181. (kafka.zookeeper.ZooKeeperClient)
[2026-04-25 21:03:29,724] INFO Client environment:zookeeper.version=3.8.4-9316c2a7a97a1e66d8f4593f34dd6fc36ec436c, built on 2024-02-12 22:16 UTC (org.apache.zookeeper.ZooKeeper)

[root@node2 config]# /export/server/kafka/bin/kafka-server-start.sh /export/server/kafka/config/server.properties
[2026-04-25 21:03:41,080] INFO Registered kafka:type=kafka.Log4jController MBean (kafka.utils.Log4jControllerRegistration$)
[2026-04-25 21:03:41,103] INFO Setting -D jdk.tls.rejectClientInitiatedRenegotiation=true to disable client-initiated TLS renegotiation (org.apache.zookeeper.common.X509Util)
[2026-04-25 21:03:45,898] INFO Registered signal handlers for TERM, INT, HUP (org.apache.kafka.common.utils.LoggingSignalHandler)
```

出现以下字样，说明服务启动成功

```
INFO [KafkaServer id=1] started (kafka.server.KafkaServer)
INFO [zk-broker-1-to-controller-alter-partition-channel-manager]: Recorded new
road)

[2026-04-25 21:03:48,659] INFO [KafkaServer id=2] started (kafka.server.KafkaServer)
[2026-04-25 21:03:48,835] INFO [zk-broker-2-to-controller-forwarding-channel-manager]: Recorded new controller
ad)

[2026-04-25 21:03:48,659] INFO [KafkaServer id=3] started (kafka.server.KafkaServer)
[2026-04-25 21:03:48,835] INFO [zk-broker-3-to-controller-alter-partition-channel-manager]: Recorded new controller
rokerToControllerRequestThread)
```

方式 2——【后台启动】:

分别在 node1、2、3 上执行 `nohup /export/server/kafka/bin/kafka-server-start.sh /export/server/kafka/config/server.properties 2>&1 >> /export/server/kafka/kafka-server.log &`

`Nohup...&`, 中间所包围的命令会放到后台执行

在 node3 上执行之后, 不会像前台启动那样产生一堆日志, Kafka 程序只会在后台默默进行

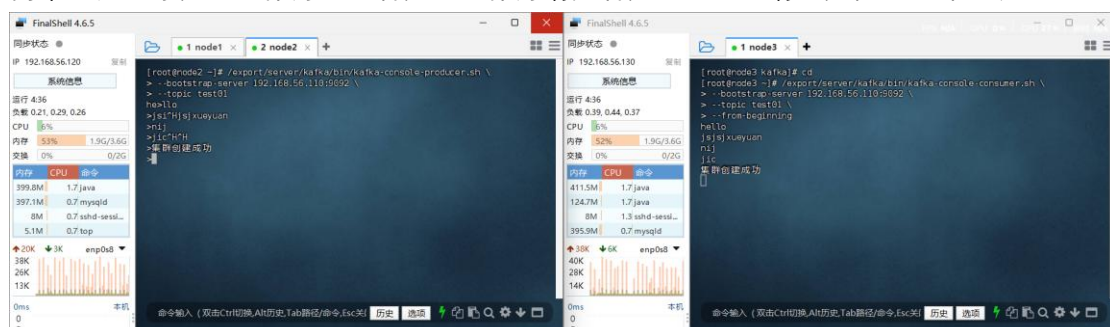
```
[root@node3 kafka]# nohup /export/server/kafka/bin/kafka-server-start.sh /export/server/kafka/config/server.properties 2>&1 >> /export/server/kafka/kafka-server.log &
[1] 951666
[root@node3 kafka]# nohup: 忽略输入重定向错误到标准输出端
```

使用 `jps` 指令: 查看所有 Java 程序

```
[root@node3 kafka]# jps
276269 QuorumPeerMain
865569 Kafka
1325160 -- process information unavailable
955024 Jps
[root@node3 kafka]#
```

可以看到 kafka 在允许状态

简单验证程序, 左边作为生产者, 右边作为消费者, 左边发送消息, 在右边都可以看到



至此 kafka 安装部署完成